Yangjia Hu

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Education

Hong Kong University of Science and Technology (HKUST) — Hong Kong
Ph.D. student in Computer Science and Engineering; Advisor: Prof. Song Guo — Sep 2026 – Jun 2030 (expected)

University of Science and Technology of China (USTC) — Hefei, China
B.Eng. in Computer Science and Technology — Sep 2022 – Jun 2026

- **GPA:** 3.91/4.3 **Rank:** 4/167

Research Interests

Efficient LLM inference, low-bit quantization, and GPU–CPU memory management.

Research Experience

MosaicQuant: Unified 4-bit LLM Quantization

HKUST; Advisor: Prof. Song Guo — Jul – Sep 2025; revised 2026

- **Proposed dense–sparse 4-bit residual compensation**: a dense base plus block-sparse residuals to correct outlier-induced error, with accuracy evaluated in INT4 and FP4.
- **Conducted the main quantization and inference experiments**. The framework selects residual blocks using activation-aware Δ-Hessian scores and fuses dense–sparse 4-bit computation through ZipperEngine.
- INT4 achieved up to **1.24× end-to-end throughput** versus TensorRT-LLM on an RTX 4090; FP4 W4A4 on Qwen3-32B attained **74.61%** average accuracy across six zero-shot tasks, versus 75.2% for W16A16.
- **Research output:** Co-first-author paper; arXiv:2606.15652 (earlier version; not yet updated).

KV Cache Prefetching for Long-context LLM Inference

Undergraduate thesis, USTC; Advisor: Jianliang Lu — Sep 2025 – Jun 2026

- Analyzed step-wise access locality and layer variation in DeepSeek-V3.2 sparse-attention traces across 61 Transformer layers; developed cacheability measures for hierarchical KV cache management.
- Designed step-locality prefetching and dual-watermark eviction, using overlap in consecutive Top-K accesses for asynchronous prefetching and synchronous miss handling.
- Built a LongBench-v2 trace-driven simulator: versus on-demand loading, per-step H2D transfers fell from **108 to 32 KV entries** (about 70%) and the miss rate from **5.28% to 1.58%**.

Related Industry Experience

Tencent — Beijing
Software Development Intern, Inference Infrastructure – Base Engine Team — Mar – Jul 2026

- Translated the thesis KV cache offloading approach into an HY4 prototype and collaborated on vLLM integration; owned asynchronous step-locality prefetching for sparse-attention decoding.
- Co-designed GPU/host cold-replica placement for 60K-token contexts under prefill/decode disaggregation. Final production validation results are pending.

Systems Project

Linux Kernel bpf_trace Module in Rust

USTC systems course project — Apr – Jul 2024

- Reworked tracing data structures and key bpf_trace routines in Rust, addressing memory safety and compatibility in kernel code.
- Integrated the migrated module into the Make build and debugged behavioral consistency with the original C implementation.

Selected Honors

Guo Moruo Scholarship (top 1%)	2026
Hong Kong Government Scholarship	2026
National Scholarship (top 2%)	2024, 2025

Technical Skills

Python, C/C++, Rust, CUDA, PyTorch, vLLM; quantization experiments, GPU/CPU performance analysis, trace-driven simulation, and KV cache management.